EE 102: Signal Processing and Linear Systems

Problem Set #9: Fourier Transform Properties

Name: _____ **Submission Date:** _____

Problem 1 Pulsating signals are all around us! The frequency domain representation of a pulse signal is quite useful. So, in this problem, consider an aperiodic pulse given below and answer the questions that follow

$$p(t) = \begin{cases} 2, & -1 < t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) [30 points] Sketch $p(t)$. Before calculating its transform, use its symmetry to predict whether $P(\omega)$ is real and whether it is even. Then derive $P(\omega)$ and give its value at $\omega = 0$.

(b) [40 points] Let $q(t) = p(t-2)$. Sketch $q(t)$ and obtain $Q(\omega)$ using the time-shifting property (see week 9 lecture 1). Explain why $|Q(\omega)| = |P(\omega)|$ even though $q(t)$ is not even about $t = 0$.

(c) [30 points] Now let $r(t) = p(t-2) + p(t+2)$. Without performing another transform integral, find $R(\omega)$ and write it as a real, even expression. Evaluate $R(0)$ and $R(\pi/4)$. Explain how the two shifted pulses can cancel at one frequency even though both pulses are positive in time.